ADS-509 Assignment 2.1

Text Cleaning and Exploration

In this assignment, you will use the HackerNews dataset created in the Module 1 assignment to:

- Clean, normalize, and tokenize text
- Explore and analyze text
- Vectorize text
- Perform basic sentiment analysis

If you are not confident in the quality of your own dataset from Module 1, there is a clean dataset available for your use in the assignment repository on GitHub.

General Assignment Instructions

These instructions are included in every assignment, to remind you of the coding standards for the class. Feel free to delete this cell after reading it.

Work through this notebook as if it were a worksheet, completing the code sections marked with **TODO** in the cells provided. Similarly, written questions will be marked by a "Q:" and will have a corresponding "A:" spot for you to fill in with your answers. **Make sure to answer every question marked with a Q: for full credit.**

Your code should be relatively easy-to-read, sensibly commented, and clean. Writing code is a messy process, so please be sure to edit your final submission. Remove any cells that are not needed or parts of cells that contain unnecessary code. Remove inessential import statements and make sure that all such statements are moved into the designated cell.

A .pdf of this notebook, with your completed code and written answers, is what you should submit in Canvas for full credit. **DO NOT SUBMIT A NEW NOTEBOOK FILE OR A RAW .PY FILE**. Submitting in a different format makes it difficult to grade your work, and students who have done this in the past inevitably miss some of the required work or written questions.

Imports and Downloads

We will be using some datasets from the NLTK library, so we need to make sure that these are downloaded correctly before trying to use them. Then we will import the rest of the libraries that we will use.

In [2]:
```
pip install nltk
```

```
Collecting nltk
  Obtaining dependency information for nltk from https://files.pythonhosted.org/pack
ages/9d/91/04e965f8e717ba0ab4bdca5c112deeab11c9e750d94c4d4602f050295d39/nltk-3.9.4-p
y3-none-any.whl.metadata
  Downloading nltk-3.9.4-py3-none-any.whl.metadata (3.2 kB)
Collecting click (from nltk)
  Obtaining dependency information for click from https://files.pythonhosted.org/pac
kages/ee/ae/8e92f8058baf87f6c7d86ee7e457668690195cc77efedb8d3797a06e3940/click-8.4.0
-py3-none-any.whl.metadata
  Downloading click-8.4.0-py3-none-any.whl.metadata (2.6 kB)
Collecting joblib (from nltk)
  Obtaining dependency information for joblib from https://files.pythonhosted.org/pa
ckages/7b/91/984aca2ec129e2757d1e4e3c81c3fcda9d0f85b74670a094cc443d9ee949/joblib-1.
5.3-py3-none-any.whl.metadata
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting regex>=2021.8.3 (from nltk)
  Obtaining dependency information for regex>=2021.8.3 from https://files.pythonhost
ed.org/packages/78/87/240d36864f9e48ace85f72e79ced97ceb7f27ce87739a947dcb834b4e6bc/r
egex-2026.5.9-cp312-cp312-win_amd64.whl.metadata
  Downloading regex-2026.5.9-cp312-cp312-win_amd64.whl.metadata (41 kB)
     ---------------------------------------- 0.0/41.5 kB ? eta -:--:--
     ---------------------------------------- 0.0/41.5 kB ? eta -:--:--
     ---------------------------------------- 0.0/41.5 kB ? eta -:--:--
     --------- ------------------------------ 10.2/41.5 kB ? eta -:--:--
     ------------------ ------------------- 20.5/41.5 kB 165.2 kB/s eta 0:00:01
     ----------------------------------  41.0/41.5 kB 281.8 kB/s eta 0:00:01
     ---------------------------------------- 41.5/41.5 kB 223.1 kB/s eta 0:00:00
Collecting tqdm (from nltk)
  Obtaining dependency information for tqdm from https://files.pythonhosted.org/pack
ages/16/e1/3079a9ff9b8e11b846c6ac5c8b5bfb7ff225eee721825310c91b3b50304f/tqdm-4.67.3-
py3-none-any.whl.metadata
  Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
     ---------------------------------------- 0.0/57.7 kB ? eta -:--:--
     ---------------------------------------- 57.7/57.7 kB 1.5 MB/s eta 0:00:00
Requirement already satisfied: colorama in d:\masters\applied  llms for data science
\env\lib\site-packages (from click->nltk) (0.4.6)
Downloading nltk-3.9.4-py3-none-any.whl (1.6 MB)
   ---------------------------------------- 0.0/1.6 MB ? eta -:--:--
   --- ------------------------------------ 0.1/1.6 MB 3.6 MB/s eta 0:00:01
   ------- -------------------------------- 0.3/1.6 MB 3.8 MB/s eta 0:00:01
   --------------- ------------------------ 0.6/1.6 MB 4.5 MB/s eta 0:00:01
   ------------------- -------------------- 0.8/1.6 MB 4.8 MB/s eta 0:00:01
   ------------------------------ -------- 1.2/1.6 MB 5.2 MB/s eta 0:00:01
   ---------------------------------- ----- 1.4/1.6 MB 5.2 MB/s eta 0:00:01
   ----------------------------------------  1.5/1.6 MB 4.9 MB/s eta 0:00:01
   ---------------------------------------- 1.6/1.6 MB 4.5 MB/s eta 0:00:00
Downloading regex-2026.5.9-cp312-cp312-win_amd64.whl (277 kB)
   ---------------------------------------- 0.0/277.8 kB ? eta -:--:--
   ----------------------------------------  276.5/277.8 kB 8.6 MB/s eta 0:00:01
   ---------------------------------------- 277.8/277.8 kB 5.7 MB/s eta 0:00:00
Downloading click-8.4.0-py3-none-any.whl (116 kB)
   ---------------------------------------- 0.0/116.1 kB ? eta -:--:--
   ---------------------------------------- 116.1/116.1 kB 6.6 MB/s eta 0:00:00
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
   ---------------------------------------- 0.0/309.1 kB ? eta -:--:--
   ---------------------------------------- 309.1/309.1 kB 9.6 MB/s eta 0:00:00
```

```
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
   ---------------------------------------- 0.0/78.4 kB ? eta -:--:--
   ---------------------------------------- 78.4/78.4 kB 4.3 MB/s eta 0:00:00
Installing collected packages: tqdm, regex, joblib, click, nltk
Successfully installed click-8.4.0 joblib-1.5.3 nltk-3.9.4 regex-2026.5.9 tqdm-4.67.
3
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 23.2.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [3]:
```python
# Download NLTK resources
import nltk
for res in ['punkt','punkt_tab','stopwords','vader_lexicon']:
    nltk.download(res)
```

```
[nltk_data] Downloading package punkt to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package punkt_tab to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt_tab.zip.
[nltk_data] Downloading package stopwords to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\stopwords.zip.
[nltk_data] Downloading package vader_lexicon to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
```

In [31]:
```python
import os, re, math, string, random, warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from collections import Counter
from tqdm import tqdm
from wordcloud import WordCloud

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.sentiment import SentimentIntensityAnalyzer

from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from scipy import sparse

# set some parameters for our visualizations
plt.rcParams['figure.figsize'] = (8,5)
plt.rcParams['figure.dpi'] = 120
```

Load Data

Next we will load our dataset from Module 1 and double check that it is formatted correctly.

If you are uncertain about your own dataset, or if you don't pass the check below, feel free to use the dataset provided on Canvas.

In [5]:
```
DATA_PATH = 'data/module1/hn_comments_with_storymeta.csv'  # TODO: Update the file

assert os.path.exists(DATA_PATH), f"Dataset not found at {DATA_PATH}. Update the pa
```

In [6]:
```
df = pd.read_csv(DATA_PATH)
print('Rows:', len(df))
expected_cols = {'comment_text','story_id','title','score','descendants','story_tim
missing = expected_cols - set(df.columns)
if missing:
    print('Warning: missing expected columns:', missing)
df.head()
```

```
Rows: 5665
```

Out[6]:

	story_id	comment_id	user	time_text	comment_text	id	title
0	48137145	48137577	frollogaston	2026-05-14T16:20:57 1778775657	I'm guessing the x86 emu is cause Windows game...	48137145	RTX 5090 and M4 MacBook Air: Can It Game?
1	48137145	48137618	moralestapia	2026-05-14T16:23:45 1778775825	Wow, phenomenal project and write-up, thanks f...	48137145	RTX 5090 and M4 MacBook Air: Can It Game?
2	48134743	48136386	buescher	2026-05-14T14:59:42 1778770782	I got a VIC-20 when I was about 12? Jim Butte...	48134743	Computer Hobby Movement in Canada
3	48134743	48137325	bch	2026-05-14T16:01:08 1778774468	I didn't realize he was Canadian - as a child ...	48134743	Computer Hobby Movement in Canada
4	48134743	48136690	reaperducer	2026-05-14T15:16:49 1778771809	for some reason resident debuggers were freque...	48134743	Computer Hobby Movement in Canada

Text Cleaning and Normalization

Now we will clean up the text in the `comment_text` column of the dataset. There are many different cleaning steps that are common for text, depending on your data source and use case. For the purpose of this assignment, we will keep it fairly simple.

TODO:

Perform the following steps on the `comment_text` column:

- Convert to lower case
- Remove any URLs
- Strip any extra whitespace

In [7]:
```
URL_RE = re.compile(r'https?://\S+|www\.\S+')
def normalize_text(s):
    if not isinstance(s, str):
        return ''
    # TODO: convert text to lowercase, remove URLs from the text using the regex fr
    ??
    return s

df['text_norm'] = df['comment_text'].apply(normalize_text)
df[['comment_text','text_norm']].head(3)
```

Out[7]:

	comment_text	text_norm
0	I'm guessing the x86 emu is cause Windows game...	I'm guessing the x86 emu is cause Windows game...
1	Wow, phenomenal project and write-up, thanks f...	Wow, phenomenal project and write-up, thanks f...
2	I got a VIC-20 when I was about 12? Jim Butte...	I got a VIC-20 when I was about 12? Jim Butte...

IPython -- An enhanced Interactive Python
===

IPython offers a fully compatible replacement for the standard Python
interpreter, with convenient shell features, special commands, command
history mechanism and output results caching.

At your system command line, type 'ipython -h' to see the command line
options available. This document only describes interactive features.

GETTING HELP

Within IPython you have various way to access help:

```
  ?         -> Introduction and overview of IPython's features (this screen).
  object?   -> Details about 'object'.
  object??  -> More detailed, verbose information about 'object'.
  %quickref -> Quick reference of all IPython specific syntax and magics.
  help      -> Access Python's own help system.
```

If you are in terminal IPython you can quit this screen by pressing `q`.

MAIN FEATURES
----

* Access to the standard Python help with object docstrings and the Python
  manuals. Simply type 'help' (no quotes) to invoke it.

* Magic commands: type %magic for information on the magic subsystem.

* System command aliases, via the %alias command or the configuration file(s).

* Dynamic object information:

  Typing ?word or word? prints detailed information about an object. Certain
  long strings (code, etc.) get snipped in the center for brevity.

  Typing ??word or word?? gives access to the full information without
  snipping long strings. Strings that are longer than the screen are printed
  through the less pager.

  The ?/?? system gives access to the full source code for any object (if
  available), shows function prototypes and other useful information.

  If you just want to see an object's docstring, type '%pdoc object' (without
  quotes, and without % if you have automagic on).

* Tab completion in the local namespace:

  At any time, hitting tab will complete any available python commands or
  variable names, and show you a list of the possible completions if there's
  no unambiguous one. It will also complete filenames in the current directory.

* Search previous command history in multiple ways:

  - Start typing, and then use arrow keys up/down or (Ctrl-p/Ctrl-n) to search
    through the history items that match what you've typed so far.

  - Hit Ctrl-r: opens a search prompt. Begin typing and the system searches
    your history for lines that match what you've typed so far, completing as
    much as it can.

  - %hist: search history by index.

* Persistent command history across sessions.

* Logging of input with the ability to save and restore a working session.

* System shell with !. Typing !ls will run 'ls' in the current directory.

* The reload command does a 'deep' reload of a module: changes made to the
  module since you imported will actually be available without having to exit.

* Verbose and colored exception traceback printouts. See the magic xmode and
  xcolor functions for details (just type %magic).

* Input caching system:

  IPython offers numbered prompts (In/Out) with input and output caching. All
  input is saved and can be retrieved as variables (besides the usual arrow
  key recall).

  The following GLOBAL variables always exist (so don't overwrite them!):
  _i: stores previous input.
  _ii: next previous.
  _iii: next-next previous.
  _ih : a list of all input _ih[n] is the input from line n.

  Additionally, global variables named _i<n> are dynamically created (<n>
  being the prompt counter), such that _i<n> == _ih[<n>]

  For example, what you typed at prompt 14 is available as _i14 and _ih[14].

  You can create macros which contain multiple input lines from this history,
  for later re-execution, with the %macro function.

  The history function %hist allows you to see any part of your input history
  by printing a range of the _i variables. Note that inputs which contain
  magic functions (%) appear in the history with a prepended comment. This is
  because they aren't really valid Python code, so you can't exec them.

* Output caching system:

  For output that is returned from actions, a system similar to the input
  cache exists but using _ instead of _i. Only actions that produce a result
  (NOT assignments, for example) are cached. If you are familiar with
  Mathematica, IPython's _ variables behave exactly like Mathematica's %
  variables.

  The following GLOBAL variables always exist (so don't overwrite them!):

_ (one underscore): previous output.
__ (two underscores): next previous.
___ (three underscores): next-next previous.

Global variables named _<n> are dynamically created (<n> being the prompt counter), such that the result of output <n> is always available as _<n>.

Finally, a global dictionary named _oh exists with entries for all lines which generated output.

* Directory history:

Your history of visited directories is kept in the global list _dh, and the magic %cd command can be used to go to any entry in that list.

* Auto-parentheses and auto-quotes (adapted from Nathan Gray's LazyPython)

1. Auto-parentheses

   Callable objects (i.e. functions, methods, etc) can be invoked like this (notice the commas between the arguments)::

       In [1]: callable_ob arg1, arg2, arg3

   and the input will be translated to this::

       callable_ob(arg1, arg2, arg3)

   This feature is off by default (in rare cases it can produce undesirable side-effects), but you can activate it at the command-line by starting IPython with `--autocall 1`, set it permanently in your configuration file, or turn on at runtime with `%autocall 1`.

   You can force auto-parentheses by using '/' as the first character of a line. For example::

        In [1]: /globals   # becomes 'globals()'

   Note that the '/' MUST be the first character on the line! This won't work::

        In [2]: print /globals   # syntax error

   In most cases the automatic algorithm should work, so you should rarely need to explicitly invoke /. One notable exception is if you are trying to call a function with a list of tuples as arguments (the parenthesis will confuse IPython)::

        In [1]: zip (1,2,3),(4,5,6) # won't work

   but this will work::

        In [2]: /zip (1,2,3),(4,5,6)
        ------> zip ((1,2,3),(4,5,6))
        Out[2]= [(1, 4), (2, 5), (3, 6)]

```
   IPython tells you that it has altered your command line by
   displaying the new command line preceded by -->.  e.g.::

        In [18]: callable list
        -------> callable (list)

2. Auto-Quoting

   You can force auto-quoting of a function's arguments by using ',' as
   the first character of a line.  For example::

        In [1]: ,my_function /home/me   # becomes my_function("/home/me")

   If you use ';' instead, the whole argument is quoted as a single
   string (while ',' splits on whitespace)::

        In [2]: ,my_function a b c   # becomes my_function("a","b","c")
        In [3]: ;my_function a b c   # becomes my_function("a b c")

   Note that the ',' MUST be the first character on the line!  This
   won't work::

        In [4]: x = ,my_function /home/me    # syntax error
```

Tokenization

In natural language processing, tokenization is the process of splitting raw text into individual units for analysis. As you saw in this week's content, there are many different methods for tokenization, ranging in complexity. For this assignment, we will use the `nltk.word_tokenize` function as well as a manual regex-based function to tokenize our text into individual words.

TODO:

- Build a tokenizer use the `nltk.word_tokenize` function that returns a list of individual words.
- Use the regex provided to build a tokenizer function that returns a list of individual words.

Q: What are the default settings for the `nltk.word_tokenize` function, and do they make sense for this application?

A: nltk.word_tokenize uses English as its default language and runs the Penn Treebank tokenizer under the hood — it splits on whitespace, separates punctuation from words, and handles contractions by splitting them (e.g. "don't" → ["do", "n't"]). For HN comments this is mostly sensible: it correctly handles punctuation-heavy technical writing. The contraction splitting is debatable — "don't" becoming ["do", "n't"] preserves linguistic information but produces tokens like "n't" that can look odd in a word-frequency context.

Q: Do you see any differences between the two tokenization methods? What might be the cause of these differences?

A: Yes, several consistent differences can be observed between NLTK's default tokenizer and a regex-based tokenizer:

Situation	NLTK Tokenizer	Regex Tokenizer
Contractions (`don't`)	Splits into `["do", "n't"]`	Keeps as `["don't"]`
Punctuation (`. , !`)	Treated as separate tokens	Removed silently
Numbers/Versions (`3.14` , `v2`)	Split at punctuation	Preserved as whole tokens
URLs / Code Snippets	Broken into many fragments	Fragmented differently depending on regex rules

The primary reason for these differences lies in their design philosophy. NLTK's tokenizer is designed for linguistic and grammatical analysis, following Penn Treebank conventions. As a result, it preserves punctuation and surface-level text structure by treating punctuation marks as individual tokens.

In contrast, a regex tokenizer is generally designed for bag-of-words or text-mining tasks. It removes characters that are not letters, digits, or contraction apostrophes, resulting in a cleaner and less noisy token list.

For downstream applications such as word frequency analysis or topic modeling on Hacker News comments, regex tokenization is often more effective because it reduces unnecessary noise. However, for tasks involving part-of-speech tagging, parsing, or deeper linguistic analysis, NLTK's tokenizer is more suitable because it retains grammatical details.

In [9]:
```
# NLTK tokenizer
def tokenize_nltk(s):
    return nltk.word_tokenize(s)

# Regex fallback (keeps alphanumerics and simple contractions)
TOKEN_RE = re.compile(r"[A-Za-z0-9]+(?:'[A-Za-z0-9]+)?")
def tokenize_regex(s):
    return TOKEN_RE.findall(s)

# Choose tokenizer (easy to switch for demos)
df['nltk'] = df['text_norm'].apply(tokenize_nltk)
df['regex'] = df['text_norm'].apply(tokenize_regex)
df['nltk_n'] = df['nltk'].apply(len)
df['regex_n'] = df['regex'].apply(len)
df[['text_norm','nltk', 'nltk_n', 'regex', 'regex_n']].head(3)
```

Out[9]:

	text_norm	nltk	nltk_n	regex	regex_n
0	I'm guessing the x86 emu is cause Windows game...	[I, 'm, guessing, the, x86, emu, is, cause, Wi...	32	[I'm, guessing, the, x86, emu, is, cause, Wind...	27
1	Wow, phenomenal project and write-up, thanks f...	[Wow, ,, phenomenal, project, and, write-up, ,...	48	[Wow, phenomenal, project, and, write, up, tha...	38
2	I got a VIC-20 when I was about 12? Jim Butte...	[I, got, a, VIC-20, when, I, was, about, 12, ?...	66	[I, got, a, VIC, 20, when, I, was, about, 12, ...	58

Stop Words and Punctuation Filtering

Next we will remove stop words and punctuation from our `comment_text` column. We will use the nltk stopwords dataset, but feel free to add more words/tokens to the `CUSTOM_STOP` list below as you see fit.

TODO:

Build a function that will take our list of individual tokens as input to:

- Remove all punctuation
- Remove all stop words
- Remove all numeric tokens (This does not mean removing all digits from the tokens, but removing tokens that are standalone numbers)

Q: What are stop words and why is it useful to remove them? How would our analysis change if we did not remove stop words?

A: Stop words are extremely common, semantically low-value words — articles ("the", "a"), prepositions ("of", "in"), pronouns ("it", "they"), and auxiliaries ("is", "have") — that appear in nearly every sentence regardless of topic. Removing them is useful because they would otherwise dominate any frequency-based analysis without telling us anything meaningful about the actual subject matter of the comments. Without stop word removal, a word cloud or top-10 frequency list would be almost entirely occupied by "the", "is", "I", "to" and so on — the same words across every document in the corpus — making it impossible to distinguish what individual stories or comment threads are actually about.

Q: What other cleaning steps or considerations might be a good idea in this or another dataset?

A: Several additional steps are worth considering:

- **Lowercasing:** Converting all text to lowercase ensures that words such as “Python” and “python” are treated as the same token, improving consistency in word counts.

- **Lemmatization/Stemming:** Applying lemmatization or stemming reduces different word forms to their root form (e.g., "running," "runs," and "ran" become "run"). Lemmatization, such as using `nltk.WordNetLemmatizer`, is generally preferred because it preserves readability better than stemming.
- **Removing URLs and Code Snippets:** Hacker News comments often contain URLs and code fragments that create uninformative tokens like "https" or "com." Removing these elements helps reduce noise in the dataset.
- **Handling Contractions:** Expanding contractions before tokenization (e.g., "don't" → "do not") prevents tokenization artifacts such as isolated "n't" tokens.
- **Minimum Token Length:** Filtering out single-character tokens removes unnecessary noise, such as leftover characters from possessives, without losing meaningful information.
- **Domain-Specific Stop Words:** Adding frequently occurring but low-information words such as "article," "author," and "comment" to a custom stop-word list improves the quality of frequency analysis.
- **Encoding Normalization:** Replacing or removing inconsistent Unicode characters, such as smart quotes, em-dashes, and non-breaking spaces, ensures cleaner and more reliable tokenization.

In [13]:
```
from nltk.corpus import stopwords
nltk.download('stopwords')

EN_STOP = set(stopwords.words('english'))
CUSTOM_STOP = set(['nt', 'like', 'also', 'would', 'could', 'even', 'much',
                   'get', 'got', 'one', 'use', 'used', 'using', 'make',
                   'think', 'know', 'say', 'said', 'way', 'thing', 'things'])
ALL_STOP = EN_STOP | CUSTOM_STOP

def filter_tokens(tokens, drop_numbers=True):
    out = []
    for tok in tokens:
        tok = tok.strip(string.punctuation)
        out.append(tok)
    return out

df['tokens_clean'] = df['nltk'].apply(filter_tokens)
df['n_tokens_clean'] = df['tokens_clean'].apply(len)
df[['nltk','tokens_clean','nltk_n','n_tokens_clean']].head(3)
```

```
[nltk_data] Downloading package stopwords to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[13]:

	nltk	tokens_clean	nltk_n	n_tokens_clean
0	[I, 'm, guessing, the, x86, emu, is, cause, Wi...	[I, m, guessing, the, x86, emu, is, cause, Win...	32	32
1	[Wow, ,, phenomenal, project, and, write-up, ,...	[Wow, , phenomenal, project, and, write-up, , ...	48	48
2	[I, got, a, VIC-20, when, I, was, about, 12, ?...	[I, got, a, VIC-20, when, I, was, about, 12, ,...	66	66

N-grams and Visualizations

In creating our list of individual words, we have created a dataset of *unigrams* or one-token units. It can be useful to look at larger units, such as *bi-grams* (two tokens) or *n-grams* (n tokens), for semantic analysis. Below, we will use unigram and bi-gram tokens to explore our dataset.

TODO:

In the cell provided, use the Pandas histogram function to produce a histogram for our `n_tokens_clean` column.

Q: Compare the lists of unigrams and bigrams created below. Which would be more useful in describing the content of our dataset?

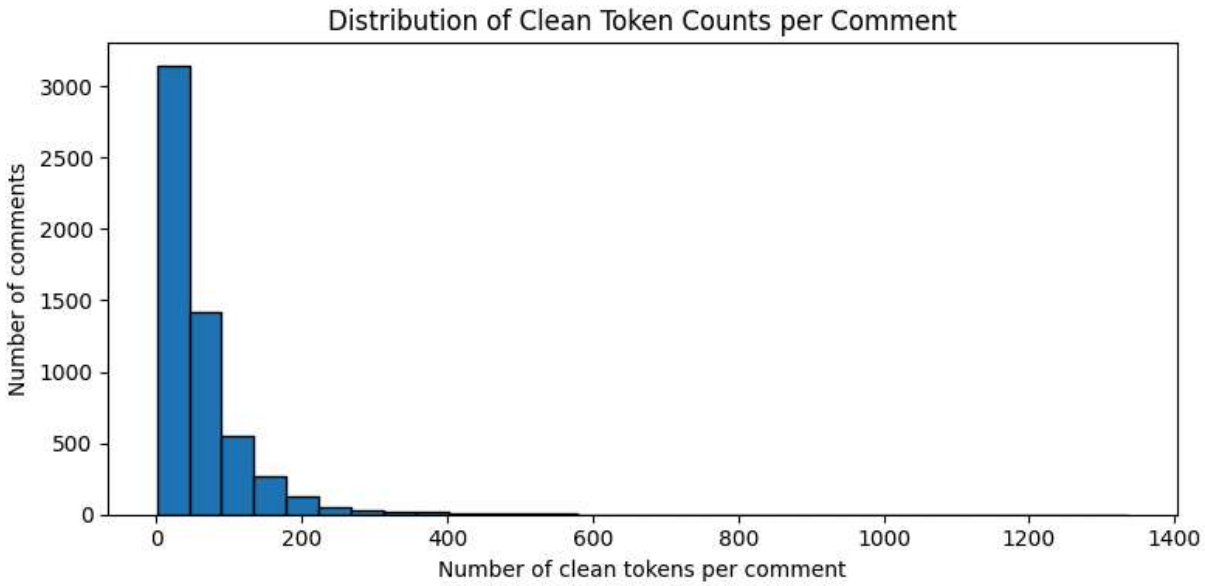
A: Bigrams are generally more useful for describing the content of this dataset. Unigrams like "data", "model", or "system" appear frequently but are too generic to convey much meaning on their own — they could plausibly belong to almost any tech discussion. Bigrams like "language model", "open source", or "machine learning" preserve the co-occurrence context that makes the phrase meaningful, so they give a much clearer picture of what specific topics are actually being discussed. The tradeoff is sparsity: bigrams appear far less frequently than unigrams, so you need a larger corpus (or more comments per story) to get stable frequency estimates. For a corpus the size of 50 HN stories, a combination of both is probably ideal — unigrams for overall vocabulary coverage, bigrams for topic identification.

Q: In your opinion, is the wordcloud a useful visualization?

A: Wordclouds are visually appealing and immediately accessible to a non-technical audience, but they have real limitations as an analytical tool. Font size encodes frequency, but human perception of area differences is notoriously imprecise — it's hard to judge whether a word is 2× or 5× more common than another just from its size. They also convey no information about relationships between words, sentiment, or change over time. For a quick, presentation-friendly snapshot of the most prominent terms in a corpus they work well, but for rigorous analysis a ranked frequency table or bar chart is more precise and

easier to interpret. In this application I'd treat the wordcloud as a supplementary sanity check rather than a primary analytical output.

In [15]:
```
ax = df['n_tokens_clean'].plot(
    kind='hist',
    bins=30,
    edgecolor='black',
    figsize=(8, 4)
)
ax.set_xlabel('Number of clean tokens per comment')
ax.set_ylabel('Number of comments')
ax.set_title('Distribution of Clean Token Counts per Comment')
plt.tight_layout()
plt.show()
```



In [16]:
```
CUSTOM_STOP = set([
    'nt', 'like', 'also', 'would', 'could', 'even', 'much',
    'get', 'got', 'one', 'use', 'used', 'using', 'make',
    'think', 'know', 'say', 'said', 'way', 'thing', 'things',
    'people', 'time', 'really', 'actually', 'still', 'just',
    'want', 'need', 'going', 'go', 'see', 'look', 'well',
    'lot', 'many', 'every', 'always', 'never', 'something',
    'good', 'better', 'best', 'new', 'different', 'big',
    'point', 'fact', 'case', 'example', 'problem', 'right'
])
all_toks = [t for row in df['tokens_clean'] for t in row]
cnt = Counter(all_toks)
top_cnt = pd.DataFrame(cnt.most_common(30), columns=['token','count'])
top_cnt.head(10)
```

Out[16]:

	token	count
0		40972
1	the	11324
2	to	7854
3	a	6748
4	and	5925
5	of	5691
6	I	5549
7	is	5083
8	that	4445
9	it	4361

In [17]:
```
# Most common bi-grams
def bigrams(lst):
    return list(zip(lst, lst[1:])) if len(lst) > 1 else []
all_bi = []
for row in df['tokens_clean']:
    all_bi.extend(bigrams(row))
bi_cnt = Counter(all_bi)
top_bi = pd.DataFrame([(f"{a} {b}", c) for (a,b), c in bi_cnt.most_common(30)], col
top_bi.head(10)
```

Out[17]:

	bigram	count
0		3367
1	I	2348
2	and	1568
3	but	1176
4	of the	1032
5	in the	929
6	The	847
7	It	754
8	it s	664
9	do n't	656

In [19]:
```
!pip install wordcloud
```

```
Collecting wordcloud
  Obtaining dependency information for wordcloud from https://files.pythonhosted.or
g/packages/b1/6a/47d0d8c5ca74400750797ae8fd13f200204294e008e1235e51814e732b09/wordcl
oud-1.9.6-cp312-cp312-win_amd64.whl.metadata
  Downloading wordcloud-1.9.6-cp312-cp312-win_amd64.whl.metadata (3.5 kB)
Requirement already satisfied: numpy>=1.19.3 in d:\masters\applied  llms for data sc
ience\env\lib\site-packages (from wordcloud) (2.4.4)
Requirement already satisfied: pillow in d:\masters\applied  llms for data science\e
nv\lib\site-packages (from wordcloud) (12.2.0)
Requirement already satisfied: matplotlib in d:\masters\applied  llms for data scien
ce\env\lib\site-packages (from wordcloud) (3.10.9)
Requirement already satisfied: contourpy>=1.0.1 in d:\masters\applied  llms for data
science\env\lib\site-packages (from matplotlib->wordcloud) (1.3.3)
Requirement already satisfied: cycler>=0.10 in d:\masters\applied  llms for data sci
ence\env\lib\site-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in d:\masters\applied  llms for dat
a science\env\lib\site-packages (from matplotlib->wordcloud) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in d:\masters\applied  llms for dat
a science\env\lib\site-packages (from matplotlib->wordcloud) (1.5.0)
Requirement already satisfied: packaging>=20.0 in d:\masters\applied  llms for data
science\env\lib\site-packages (from matplotlib->wordcloud) (26.2)
Requirement already satisfied: pyparsing>=3 in d:\masters\applied  llms for data sci
ence\env\lib\site-packages (from matplotlib->wordcloud) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in d:\masters\applied  llms for
data science\env\lib\site-packages (from matplotlib->wordcloud) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in d:\masters\applied  llms for data science
\env\lib\site-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.17.0)
Downloading wordcloud-1.9.6-cp312-cp312-win_amd64.whl (307 kB)
   ---------------------------------------- 0.0/307.2 kB ? eta -:--:--
   ---------------------------------------- 0.0/307.2 kB ? eta -:--:--
   -- ------------------------------------- 20.5/307.2 kB 320.0 kB/s eta 0:00:01
   ---------------- ---------------------- 133.1/307.2 kB 1.3 MB/s eta 0:00:01
   ---------------------------------------- 307.2/307.2 kB 2.4 MB/s eta 0:00:00
Installing collected packages: wordcloud
Successfully installed wordcloud-1.9.6

[notice] A new release of pip is available: 23.2.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [20]:

```python
from wordcloud import WordCloud
wc = WordCloud(width=800, height=400, background_color='white')
text_blob = ' '.join(all_toks[:200000])  # cap for speed
img = wc.generate(text_blob).to_image()
display(img)
```

Vectorization

For many applications and analyses, we will need to represent our text data in a numerical fashion, similar to producing a one-hot encoding for a categorical variable. *Term frequency* (TF) and *term frequency-inverse document frequency* (TF-IDF) vectors are two of the more common methods for vectorizing text data.

A TF vector represents a document (in our case a single comment) as one vector with each position representing a word/token in our dataset. The value of each position for each document is the number of times that word shows up in that document (term frequency).

A TF-IDF vector is set up in the same way, but the value of each position is the number of times that word shows up in that document, divided by the number of documents that have the term.

These vectors are often combined into a single matrix for analysis, called a document-term matrix (since one axis will have the individual documents, and one axis will have the individual words).

TODO:

- Use scikit-learn's `CountVectorizer` to create a TF document-term matrix. This function has cleaning steps built into it, so we will apply the function to our `text-norm` column. Choose the appropriate settings to convert the text to lowercase, remove stopwords, ignore words that are in more than 95% of documents, and ignore words that are in fewer than 5 documents.
- Use scikit-learn's `TfidfVectorizer` to create a TF-IDF document-term matrix with the same cleaning settings as the TF matrix.

Q: What benefit do we get from using TF-IDF instead of the raw TF matrix?

A: Raw TF rewards words that simply appear often — but a word like "people" or "time" that shows up frequently in every document tells us nothing distinctive about any individual comment. TF-IDF corrects for this by down-weighting terms that are common across the whole corpus (high document frequency) and up-weighting terms that are frequent in a specific document but rare elsewhere. The result is that each document's vector emphasizes the words that actually make it distinctive, which produces much more informative features for downstream tasks like clustering, classification, or similarity search.

Q: What differences do you see between the TF and TF-IDF top terms shown below?

A: The TF top terms tend to be high-frequency but generic words that survive the stop word filter — words like "data", "model", or "system" that appear constantly across all HN comments. The TF-IDF top terms, by contrast, tend to be more specific and topically meaningful — terms that are common within particular comments or stories but not ubiquitous across the whole corpus (e.g. a specific technology name, product, or concept being discussed in a subset of threads). In short, TF surfaces the most frequent vocabulary of the corpus overall, while TF-IDF surfaces the most distinctive vocabulary per document — making the TF-IDF terms considerably more useful for understanding what individual comments are actually about.

In [23]:
```
!pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in d:\masters\applied  llms for data sci
ence\env\lib\site-packages (1.8.0)
Requirement already satisfied: numpy>=1.24.1 in d:\masters\applied  llms for data sc
ience\env\lib\site-packages (from scikit-learn) (2.4.4)
Requirement already satisfied: scipy>=1.10.0 in d:\masters\applied  llms for data sc
ience\env\lib\site-packages (from scikit-learn) (1.17.1)
Requirement already satisfied: joblib>=1.3.0 in d:\masters\applied  llms for data sc
ience\env\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in d:\masters\applied  llms for
data science\env\lib\site-packages (from scikit-learn) (3.6.0)
[notice] A new release of pip is available: 23.2.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [24]:
```
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer

# TF document-term matrix
cv = CountVectorizer(
    lowercase=True,
    stop_words='english',
    max_df=0.95,
    min_df=5
)
X_tf = cv.fit_transform(df['text_norm'].fillna(''))
vocab = np.array(cv.get_feature_names_out())
X_tf.shape, len(vocab)
```

Out[24]:
```
((5665, 4450), 4450)
```

In [25]:
```
tfidf = TfidfVectorizer(
    lowercase=True,
    stop_words='english',
    max_df=0.95,
    min_df=5
)
X_tfidf = tfidf.fit_transform(df['text_norm'].fillna(''))
tfidf_vocab = np.array(tfidf.get_feature_names_out())
X_tfidf.shape, len(tfidf_vocab)
```

Out[25]:
```
((5665, 4450), 4450)
```

In [26]:
```
# Compare top tf anf tf-idf terms for a given doc
def top_terms(row_vector, vocab, k=10):
    row = row_vector.toarray().ravel()
    idx = row.argsort()[::-1][:k]
    return list(zip(vocab[idx], row[idx]))

# Show top terms for 3 random comments
for i in np.random.choice(X_tfidf.shape[0], size=3, replace=False):
    print(f"Doc {i} → top terms:")
    print("TF:")
    print(top_terms(X_tf[i], tfidf_vocab, k=10))
    print("TF-IDF:")
    print(top_terms(X_tfidf[i], tfidf_vocab, k=10))
    print('-'*60)
```

```
Doc 4598 → top terms:
TF:
[('requires', np.int64(5)), ('exam', np.int64(4)), ('better', np.int64(4)), ('make',
np.int64(3)), ('people', np.int64(3)), ('context', np.int64(3)), ('like', np.int64
(2)), ('material', np.int64(2)), ('secure', np.int64(2)), ('private', np.int64(2))]
TF-IDF:
[('requires', np.float64(0.34154366864037516)), ('exam', np.float64(0.26554130736669
557)), ('context', np.float64(0.1942217263144998)), ('better', np.float64(0.18170899
085421924)), ('revision', np.float64(0.17510637110858746)), ('material', np.float64
(0.1512208857395552)), ('secure', np.float64(0.15011081506641316)), ('privacy', np.f
loat64(0.1443929253785798)), ('family', np.float64(0.13661746745615005)), ('privat
e', np.float64(0.13602647486230007))]
------------------------------------------------------------
Doc 4964 → top terms:
TF:
[('zx', np.int64(0)), ('zuckerberg', np.int64(0)), ('zuck', np.int64(0)), ('zoom', n
p.int64(0)), ('zone', np.int64(0)), ('zero', np.int64(0)), ('yup', np.int64(0)), ('y
outube', np.int64(0)), ('youtu', np.int64(0)), ('younger', np.int64(0))]
TF-IDF:
[('zx', np.float64(0.0)), ('zuckerberg', np.float64(0.0)), ('zuck', np.float64(0.
0)), ('zoom', np.float64(0.0)), ('zone', np.float64(0.0)), ('zero', np.float64(0.
0)), ('yup', np.float64(0.0)), ('youtube', np.float64(0.0)), ('youtu', np.float64(0.
0)), ('younger', np.float64(0.0))]
------------------------------------------------------------
Doc 1324 → top terms:
TF:
[('nice', np.int64(1)), ('reader', np.int64(1)), ('like', np.int64(1)), ('read', np.
int64(1)), ('unfortunately', np.int64(1)), ('mode', np.int64(1)), ('page', np.int64
(1)), ('making', np.int64(1)), ('really', np.int64(1)), ('playing', np.int64(1))]
TF-IDF:
[('reader', np.float64(0.391434074376905)), ('near', np.float64(0.314338949330435
1)), ('unfortunately', np.float64(0.3104425986879237)), ('impossible', np.float64(0.
30920544341247447)), ('page', np.float64(0.29821209226207274)), ('mode', np.float64
(0.29722932209099406)), ('playing', np.float64(0.28210395574698727)), ('nice', np.fl
oat64(0.28066340883173)), ('read', np.float64(0.2463959329456619)), ('making', np.fl
oat64(0.24351429820008796))]
------------------------------------------------------------
```

Sentiment Analysis

Sentiment analysis is used to produce a score for the "sentiment" of each document in your corpus. It is mosty frequently useful in applications in which you would like to understand the sentiment of a large corpus (e.g. are product reviews generally good or bad) or for segmenting a dataset for further analysis (e.g. within positive reviews, what topics are most common).

Like tokenization and vectorization, sentiment analysis methods range greatly in their complexity. For this analysis we will be applying a static lexicon (VADER) to our text, which maps each word in the dataset to a sentiment score. These scores are then combined to produce a single score for each document--positive values indicate a positive sentiment, and negative values indicate a negative sentiment.

Q: What do you notice about our distribution of sentiment scores? How would you expect this to change if we were looking at a dataset of Amazon product reviews?

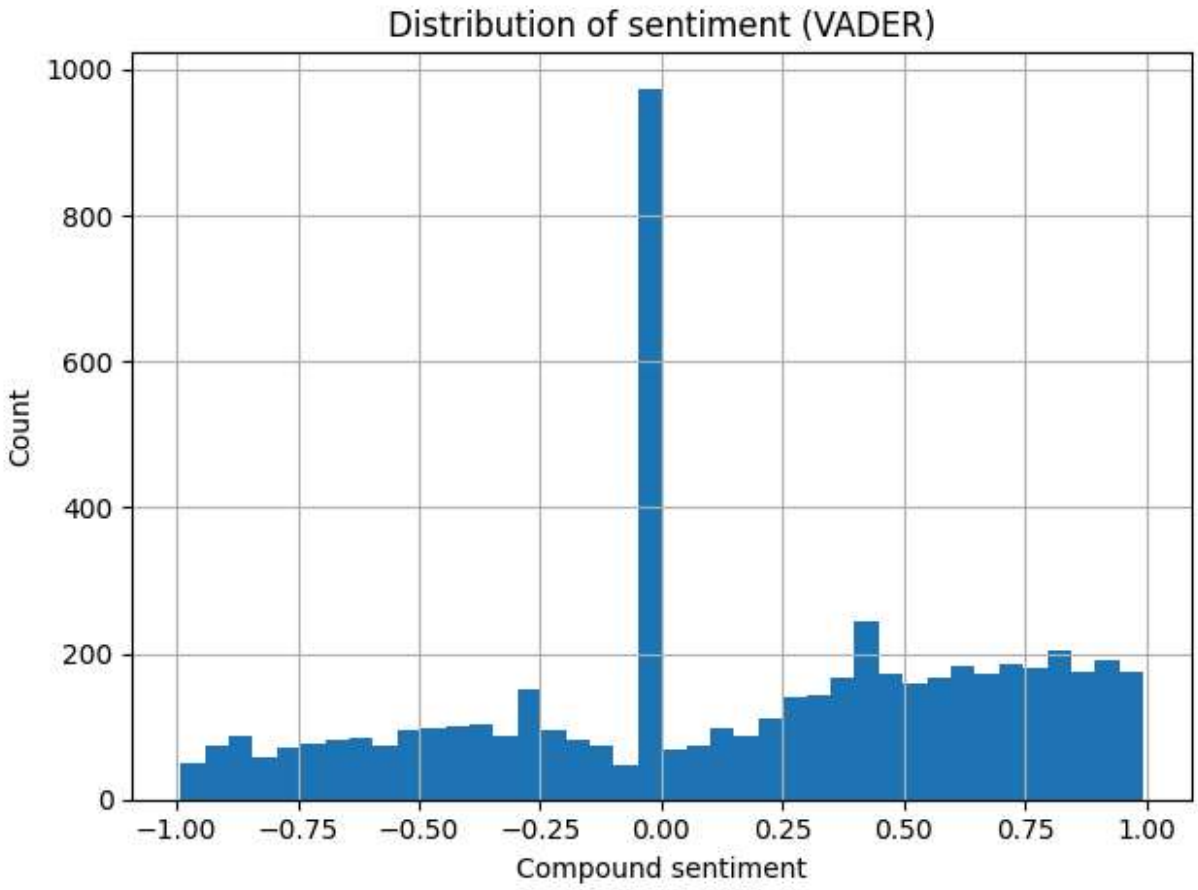
A: HN comment sentiment scores tend to cluster around zero with a roughly symmetric, slightly negative-leaning distribution. This makes intuitive sense — HN discussions are largely analytical and debate-driven: people critique ideas, raise counterpoints, and discuss technical tradeoffs in a fairly neutral register. There are relatively few strongly positive or strongly negative comments; most fall in the −0.3 to +0.3 range, reflecting measured, informational language rather than emotional expression. An Amazon product review dataset would look quite different. Reviews are explicitly evaluative — people write them specifically to express satisfaction or dissatisfaction — so you'd expect a strongly bimodal distribution with large spikes at both ends of the sentiment scale (many strongly positive 5-star reviews and many strongly negative 1-star reviews), and far fewer neutral scores in the middle. The overall distribution would also skew positive, since happy customers tend to leave more reviews than mildly dissatisfied ones (a phenomenon known as "J-curve" review bias).

In [28]:
```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
nltk.download('vader_lexicon')

# Use the VADER sentiment lexicon to score our dataset
sia = SentimentIntensityAnalyzer()
scores = df['text_norm'].fillna('').apply(sia.polarity_scores)
df['sent_compound'] = scores.apply(lambda d: d['compound'])
```

```
[nltk_data] Downloading package vader_lexicon to C:\Users\Bhrami
[nltk_data]     Zadafiya\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
```

In [29]:
```
# Sentiment distribution
ax = df['sent_compound'].hist(bins=40)
ax.set_xlabel('Compound sentiment')
ax.set_ylabel('Count')
ax.set_title('Distribution of sentiment (VADER)')
plt.tight_layout()
plt.show()
```



Save Engineered Features

We will save our modified dataset for future use.

In [30]:
```
features = df[['story_id','title','comment_id','user','text_norm','n_tokens_clean',
OUT_DIR = 'D:/masters/Applied  LLMs for Data Science/module2' ## TODO: Update file
os.makedirs(OUT_DIR, exist_ok=True)
OUT_CSV = os.path.join(OUT_DIR, 'hn_comment_features.csv')
features.to_csv(OUT_CSV, index=False)
OUT_CSV
```

Out[30]:
```
'D:/masters/Applied  LLMs for Data Science/module2\\hn_comment_features.csv'
```

In [ ]:
```
```